

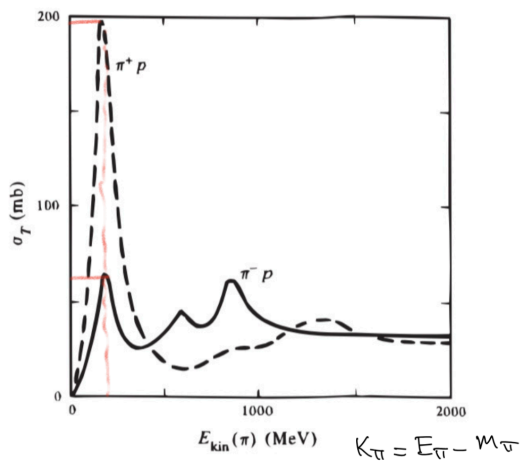
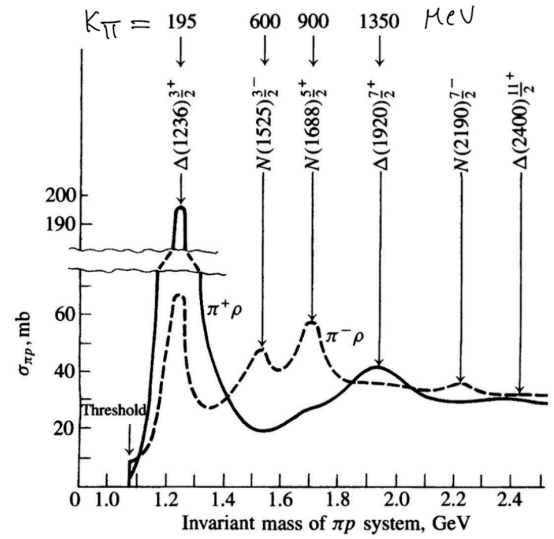
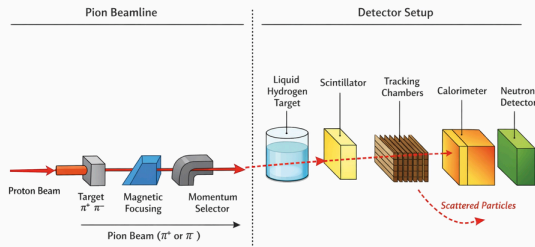


Figure 5.35: Total cross section as a function of pion kinetic energy for the scattering of positive and negative pions from protons. (1 mb = 1 millibarn = 10^{-27} cm².)



Pion-Nucleon Scattering Experiment



34. CLEBSCH-GORDAN COEFFICIENTS, SPHERICAL HARMONICS, AND d FUNCTIONS

Note: A square-root sign is to be understood over every coefficient, e.g., for $-8/15$ read $-\sqrt{8/15}$.

Notation: $\begin{matrix} j & j & j \\ m & m & m \\ M & M & M \end{matrix}$ Coefficients

$Y_l^0 = \sqrt{\frac{2}{4\pi}} \cos \theta$
 $Y_l^1 = -\sqrt{\frac{2}{4\pi}} \sin \theta e^{i\phi}$
 $Y_l^2 = \sqrt{\frac{5}{4\pi}} \left(\frac{3}{2} \cos^2 \theta - \frac{1}{2} \right)$
 $Y_l^3 = -\sqrt{\frac{15}{4\pi}} \sin \theta \cos \theta e^{i\phi}$
 $Y_l^4 = \sqrt{\frac{35}{4\pi}} \sin^2 \theta \cos^2 \theta e^{i2\phi}$

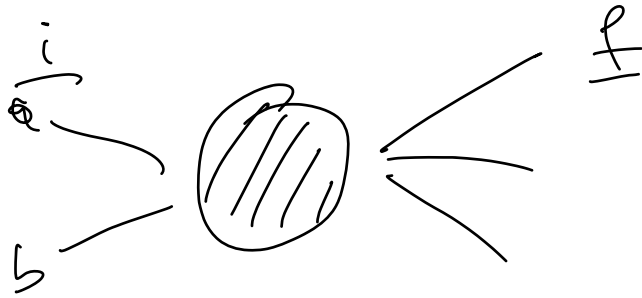
$Y_l^m = (-1)^m Y_l^{-m}$
 $d_{m,0}^l = \sqrt{\frac{4\pi}{2l+1}} Y_l^m e^{-im\phi}$

$d_{m,0}^l = \cos \theta$
 $d_{1/2,1/2}^1 = \cos \frac{\theta}{2}$
 $d_{1/2,-1/2}^1 = -\sin \frac{\theta}{2}$
 $d_{1,1}^1 = \frac{1 + \cos \theta}{2}$
 $d_{1,0}^1 = \frac{\sin \theta}{\sqrt{2}}$
 $d_{1,-1}^1 = \frac{1 - \cos \theta}{2}$

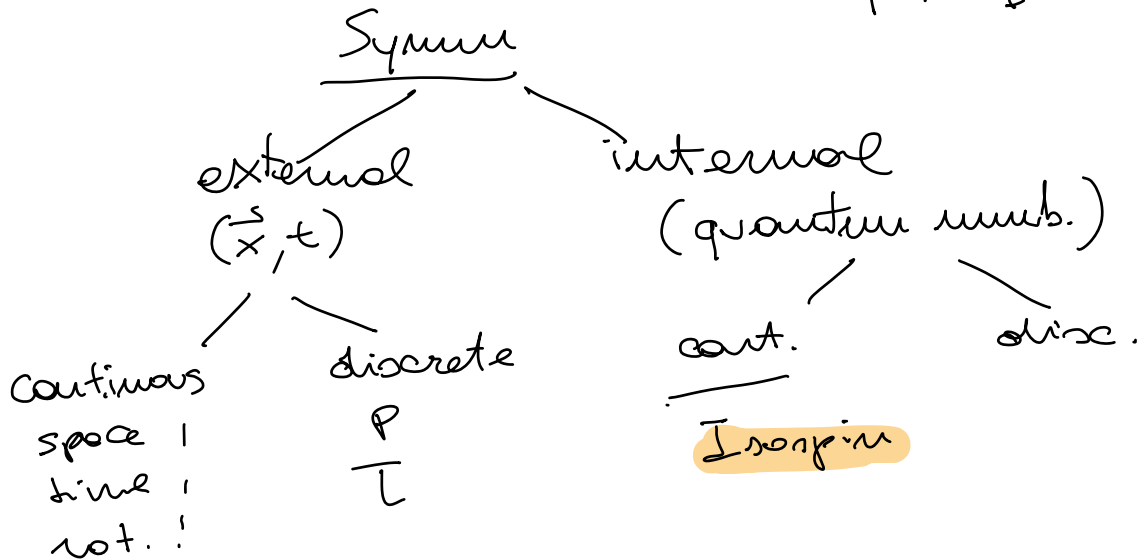
$d_{3/2,3/2}^3 = \frac{1 + \cos \theta}{2}$
 $d_{3/2,1/2}^3 = -\sqrt{\frac{1 + \cos \theta}{2}} \sin \frac{\theta}{2}$
 $d_{3/2,-1/2}^3 = \sqrt{\frac{1 - \cos \theta}{2}} \sin \frac{\theta}{2}$
 $d_{3/2,-3/2}^3 = \frac{1 - \cos \theta}{2}$
 $d_{5/2,5/2}^5 = \frac{3 \cos \theta + 1}{4}$
 $d_{5/2,3/2}^5 = \frac{3 \cos \theta - 1}{4}$
 $d_{5/2,1/2}^5 = \frac{3 \cos \theta + 1}{4}$
 $d_{5/2,-1/2}^5 = \frac{3 \cos \theta - 1}{4}$
 $d_{5/2,-3/2}^5 = \frac{3 \cos \theta + 1}{4}$
 $d_{5/2,-5/2}^5 = \frac{3 \cos \theta - 1}{4}$

$d_{1,1}^1 = \frac{1 + \cos \theta}{2}$
 $d_{1,0}^1 = \frac{\sin \theta}{\sqrt{2}}$
 $d_{1,-1}^1 = \frac{1 - \cos \theta}{2}$
 $d_{3/2,3/2}^3 = \frac{1 + \cos \theta}{2}$
 $d_{3/2,1/2}^3 = -\sqrt{\frac{1 + \cos \theta}{2}} \sin \frac{\theta}{2}$
 $d_{3/2,-1/2}^3 = \sqrt{\frac{1 - \cos \theta}{2}} \sin \frac{\theta}{2}$
 $d_{3/2,-3/2}^3 = \frac{1 - \cos \theta}{2}$
 $d_{5/2,5/2}^5 = \frac{3 \cos \theta + 1}{4}$
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 $d_{3/2,1/2}^3 = -\sqrt{\frac{1 + \cos \theta}{2}} \sin \frac{\theta}{2}$
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 $d_{5/2,1/2}^5 = \frac{3 \cos \theta + 1}{4}$
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 $d_{5/2,-3/2}^5 = \frac{3 \cos \theta + 1}{4}$
 $d_{5/2,-5/2}^5 = \frac{3 \cos \theta - 1}{4}$



events $\propto \sigma$
 $\sigma \propto |M|^2 P(E)$
 $\hookrightarrow \langle f | H_I | i \rangle$



Isospin $\Rightarrow |I, I_3\rangle$

NUCLEONS

$$p = |\frac{1}{2}, \frac{1}{2}\rangle \quad \& \quad I = \frac{1}{2}$$

$$n = |\frac{1}{2}, -\frac{1}{2}\rangle$$

PIONS

$$\pi^+ = |1, 1\rangle$$

$$\pi^0 = |1, 0\rangle \quad \& \quad I = 1$$

$$\pi^- = |1, -1\rangle$$

$$I_3 \in [-I, I]$$

↑

$$2I + 1 \text{ options}$$

Dynamical implication of I conservation

STRONG INTERACTION

$$1) NN \rightarrow d + \pi \Rightarrow 1) d = |0, 0\rangle$$

2) I is conserved
in strong inter.

$$2) \pi N \rightarrow \pi N$$

16 elastic processes

- | | |
|----------------------------------|----------------------------------|
| a) $\pi^+ p \rightarrow \pi^+ p$ | d) $\pi^+ n \rightarrow \pi^+ n$ |
| b) $\pi^0 p \rightarrow \pi^0 p$ | e) $\pi^0 n \rightarrow \pi^0 n$ |
| c) $\pi^- p \rightarrow \pi^- p$ | f) $\pi^- n \rightarrow \pi^- n$ |

4 charge exch.

- | |
|----------------------------------|
| g) $\pi^+ n \rightarrow \pi^0 p$ |
| h) $\pi^0 p \rightarrow \pi^+ n$ |
| i) $\pi^0 n \rightarrow \pi^- p$ |
| j) $\pi^- p \rightarrow \pi^0 n$ |

① produce π beam
 $p \pm A \rightarrow \pi^\pm \pi^0 X$
 $\hookrightarrow (Cu, Be, C)$

$$\tau_{\pi^0} \sim 10^{-17} s$$

$$\tau_{\pi^\pm} \sim 10^{-8} s$$

② target $\rightarrow$ Hydrogen

③ Detect outgoing particle.

a) $\pi^+ p \rightarrow \pi^+ p$

$$\pi^+ p = |1, 1\rangle \left| \frac{1}{2}, \frac{1}{2} \right\rangle$$

$$= \left| \frac{3}{2}, \frac{3}{2} \right\rangle$$

$$\sigma_a \propto |M_a|^2 \propto \left| \left\langle \frac{3}{2}, \frac{3}{2} \right| H_I \left| \frac{3}{2}, \frac{3}{2} \right\rangle \right|^2$$

$$\propto \left| M_{\frac{3}{2}} \right|^2$$

c) $\pi^- p \rightarrow \pi^- p$

$$\pi^- p = |1, -1\rangle \left| \frac{1}{2}, \frac{1}{2} \right\rangle$$

$$= \alpha \left| \frac{3}{2}, -\frac{1}{2} \right\rangle + \beta \left| \frac{1}{2}, -\frac{1}{2} \right\rangle$$

$$= \sqrt{\frac{1}{3}} \left| \frac{3}{2}, -\frac{1}{2} \right\rangle - \sqrt{\frac{2}{3}} \left| \frac{1}{2}, -\frac{1}{2} \right\rangle$$

$$\sigma_c \propto |M_c|^2$$

$$\propto \left| \left\langle -\frac{1}{2}, \frac{1}{2} \right| \left(-\sqrt{\frac{2}{3}} \right) + \left\langle -\frac{1}{2}, \frac{3}{2} \right| \left(\sqrt{\frac{1}{3}} \right) H_I \left(\sqrt{\frac{1}{3}} \left| \frac{3}{2}, -\frac{1}{2} \right\rangle - \sqrt{\frac{2}{3}} \left| \frac{1}{2}, -\frac{1}{2} \right\rangle \right) \right|^2$$

$$\propto \left| \frac{2}{3} \langle -\frac{1}{2}, \frac{1}{2} | H_I | \frac{1}{2}, -\frac{1}{2} \rangle + \frac{1}{3} \langle -\frac{1}{2}, \frac{3}{2} | H_I | \frac{3}{2}, \frac{1}{2} \rangle \right|^2$$

$$= \frac{1}{9} \left| 2 M_{\frac{1}{2}} + M_{\frac{3}{2}} \right|^2$$

i) $\pi^- p \rightarrow \pi^0 n$ $\pi^- p = \sqrt{\frac{1}{3}} | \frac{3}{2}, -\frac{1}{2} \rangle - \sqrt{\frac{2}{3}} | \frac{1}{2}, -\frac{1}{2} \rangle$

$$\pi^0 n = |1, 0\rangle | \frac{1}{2}, -\frac{1}{2} \rangle$$

$$= \sqrt{\frac{2}{3}} | \frac{3}{2}, -\frac{1}{2} \rangle + \sqrt{\frac{1}{3}} | \frac{1}{2}, -\frac{1}{2} \rangle$$

$$\sigma_j \propto |M_j|^2$$

$$M_j = \langle f | H_I | i \rangle$$

$$\propto \left| \langle -\frac{1}{2}, \frac{1}{2} | \left(-\sqrt{\frac{2}{3}} \right) + \langle -\frac{1}{2}, \frac{3}{2} | \left(\sqrt{\frac{1}{3}} \right) | H_I | \sqrt{\frac{2}{3}} | \frac{3}{2}, -\frac{1}{2} \rangle + \sqrt{\frac{1}{3}} | \frac{1}{2}, -\frac{1}{2} \rangle \right|^2$$

$$\propto \frac{2}{9} \left| M_{\frac{3}{2}} - M_{\frac{1}{2}} \right|^2$$

Recap

$$\sigma_a \propto |M_{\frac{3}{2}}|^2; \pi^+ p \rightarrow \pi^+ p$$

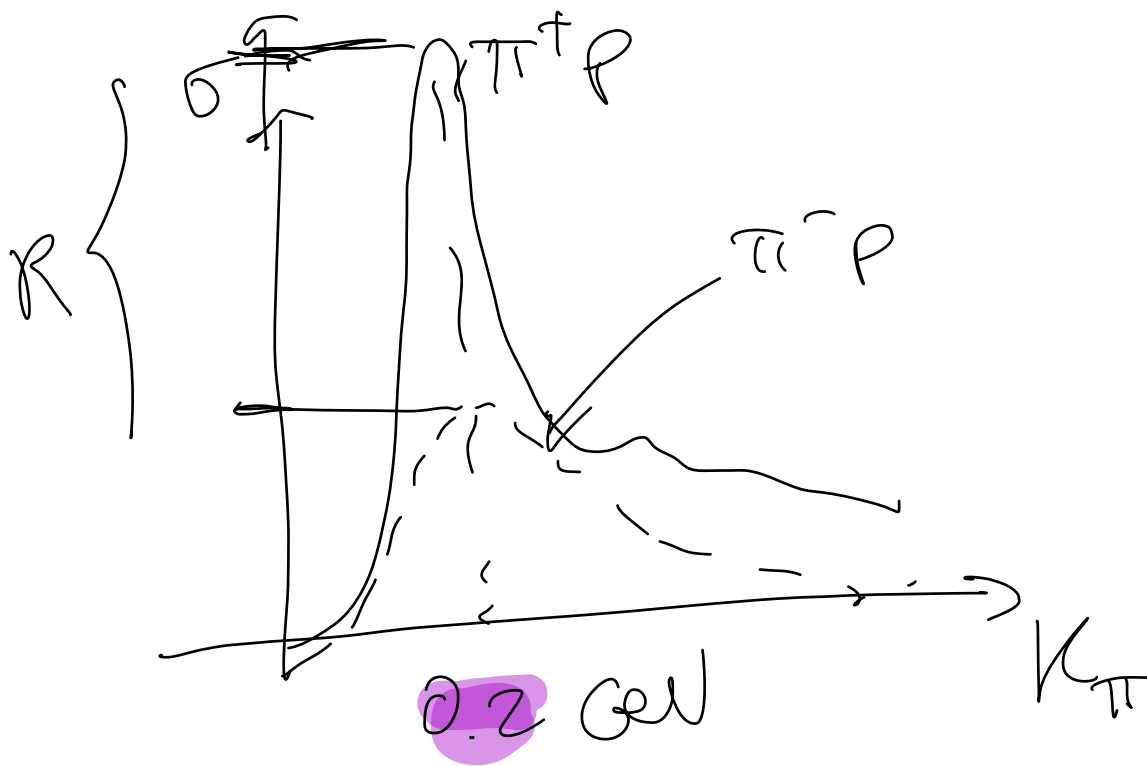
$$\sigma_c \propto \frac{1}{9} \left| 2 M_{\frac{1}{2}} + M_{\frac{3}{2}} \right|^2; \pi^+ p \rightarrow \pi^+ \bar{p}$$

$$\sigma_j \propto \frac{2}{9} \left| M_{\frac{3}{2}} - M_{\frac{1}{2}} \right|^2 \rightarrow \pi^+ p \rightarrow \pi^+ \bar{p}_n$$

$$R = \frac{\# (\pi^+ p \rightarrow \pi^+ p)}{\# (\pi^+ p \rightarrow \text{anything})}$$

$$R = \frac{\sigma_e}{\sigma_e + \sigma_j}$$

$\pi^\pm$ have $k_\pi \sim 0.2 \text{ GeV}$



$$R \sim 3$$

What does it mean

$$R = 3?$$

$$\text{if } M_{\frac{1}{2}} < M_{\frac{3}{2}} \Leftarrow$$

$\Rightarrow$ the processes are
dominated by

$$I = \frac{3}{2}$$

② What does $k_{\pi} = 0.26$
mean?



$$P_1 \begin{pmatrix} E_{\pi} \\ P_{\pi} \\ 0 \\ 0 \end{pmatrix} P_2 \begin{pmatrix} m_p \\ 1 \\ 0 \end{pmatrix}$$

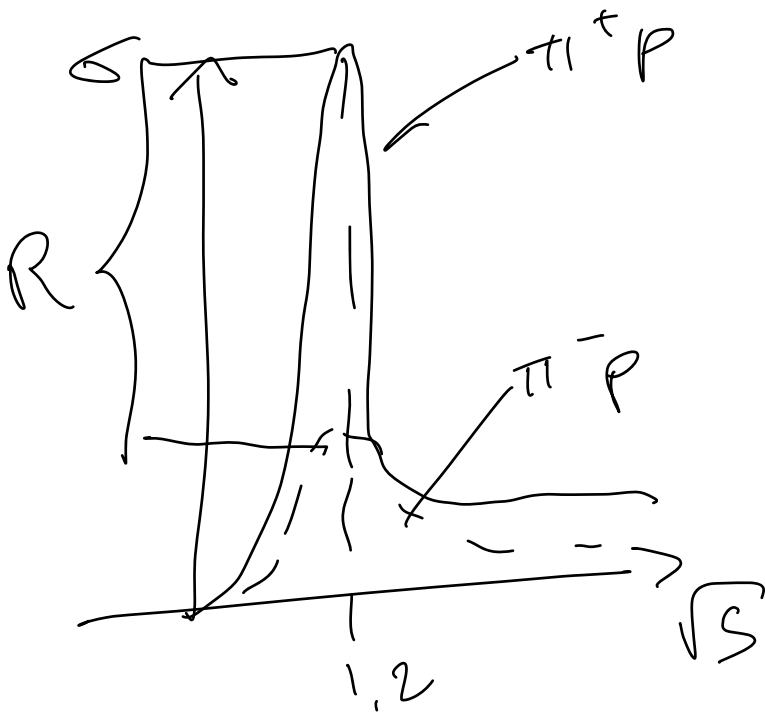
$$s = (E_\pi + m_p)^2 - |\vec{p}_\pi|^2$$

$$= E_\pi^2 + m_p^2 + 2E_\pi m_p - |\vec{p}_\pi|^2$$

$$= (m_\pi + m_p) + 2 \underbrace{E_\pi}_{(K_\pi + m_\pi)} m_p$$

$\uparrow$
0.2 GeV

$$\sqrt{s} = 1.2 \text{ GeV}$$



→ processes are domin.
by $\sqrt{s} \sim 1.2 \text{ GeV}$

$$\pi^+ p \longrightarrow \Delta^{++} \longrightarrow \pi^+ p$$



$$I = \frac{3}{2}$$

$$M = 1.2 \text{ GeV}$$